

OpenSSH Log Analysis & Splunk Security Monitoring

Comprehensive Report — Dashboard, Alert Configuration & Security Analysis

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Organization: Security Operations Team · Date: April 2026 · Version: 1.0 Final

Platform	Log Events	Tasks	Status
Splunk Enterprise 9.1.6	18 Events Indexed	5 Tasks — All Completed	✓ SUBMITTED

Task	Completion Status
Task 1 — User Management	Completed
Task 2 — Log Upload & Analysis	Completed
Task 3 — Dashboard & Alert	Completed
Task 4 — Field Extraction (src_ip)	Completed
Task 5 — Report Preparation	Completed

EXECUTIVE SUMMARY

This report documents the complete implementation of a Splunk-based security monitoring solution as part of the Security Operations team assignment. Justice Alucho successfully installed Splunk Enterprise 9.1.6 on Kali Linux, ingested an OpenSSH log file containing 18 security events, performed comprehensive threat analysis identifying suspicious IP addresses and brute-force patterns, created a live monitoring dashboard, configured automated daily alerting, performed field extraction for source IPs, and managed user accounts with appropriate role assignments. All five tasks have been completed and are documented with supporting screenshots in this submission.

1. Introduction & Assignment Context

As a newly appointed member of the Security Operations team, I — Justice Alucho — was tasked with building a comprehensive SSH log analysis and monitoring solution using Splunk. The organization had recently experienced suspicious incidents and management needed clarity on whether patterns existed in SSH login activity. This report documents every step taken, from initial Splunk installation through to dashboard creation, alert configuration, and field extraction.

The entire solution was built on Kali Linux running inside Oracle VirtualBox, with Splunk Enterprise 9.1.6 serving as the SIEM platform. All screenshots included in this report were captured live from the Splunk web interface during actual task execution.

1.1 Environment Details

Component	Details
Operating System	Kali Linux 2025.1c · VirtualBox amd64
SIEM Platform	Splunk Enterprise 9.1.6
Host IP	10.0.2.15 (VirtualBox NAT)
Splunk Web Port	http://localhost:8000
Log File	openssh_logs.log · sourcetype: syslog
Index	main
Total Events Indexed	18 events (All Time range)
Analyst	Justice Alucho · Cybersecurity Analyst

2. Task 1 — User Management

User management was the first task completed. New user accounts were created in Splunk for each group member, with specific configurations applied to each account to ensure security best practices and appropriate access controls.

2.1 User Creation Process

Each user was created by navigating to Settings → Users → New User in the Splunk web interface. The following steps were applied for every account:

1. Navigated to Settings → Users → New User.
2. Entered full name, username, and email address for each group member.
3. Set Time Zone to Africa/Lagos (WAT — West Africa Time, UTC+1).
4. Set Default App to launcher (home).
5. Checked the 'Require password change on first login' checkbox.
6. Assigned an appropriate role based on each member's responsibilities.
7. Clicked Save to create the account.

2.2 User Account Configuration Summary

Username	Role Assigned	Time Zone	Default App	Force Pwd Change
justice.alucho	admin	Africa/Lagos (WAT)	launcher (home)	Yes ✓
member2	power	Africa/Lagos (WAT)	launcher (home)	Yes ✓
member3	power	Africa/Lagos (WAT)	launcher (home)	Yes ✓
member4	user	Africa/Lagos (WAT)	launcher (home)	Yes ✓

2.3 Role Assignment Rationale

Splunk Role	Assigned To	Permissions & Rationale
admin	Team Lead / Justice Alucho	Full access to all Splunk functions including user management, index creation, and system configuration.
power	Senior Analysts	Can run searches, create dashboards, save reports and alerts. Cannot manage users or system settings.
user	Junior Members / Viewers	Read-only access. Can view dashboards and run basic searches. Cannot create alerts or modify configurations.

3. Task 2 — Log Upload & Security Analysis

3.1 Log File Upload

The OpenSSH log file (openssh_logs.log) was uploaded into Splunk via Settings → Add Data → Upload. The source type was set to 'syslog' and the data was indexed into the 'main' index. A total of 18 events were successfully ingested and made available for searching.

3.2 Evidence — Indexed Log Events (Screenshot)

The screenshot below shows the Splunk search interface with all 18 indexed events visible, confirming successful data ingestion:

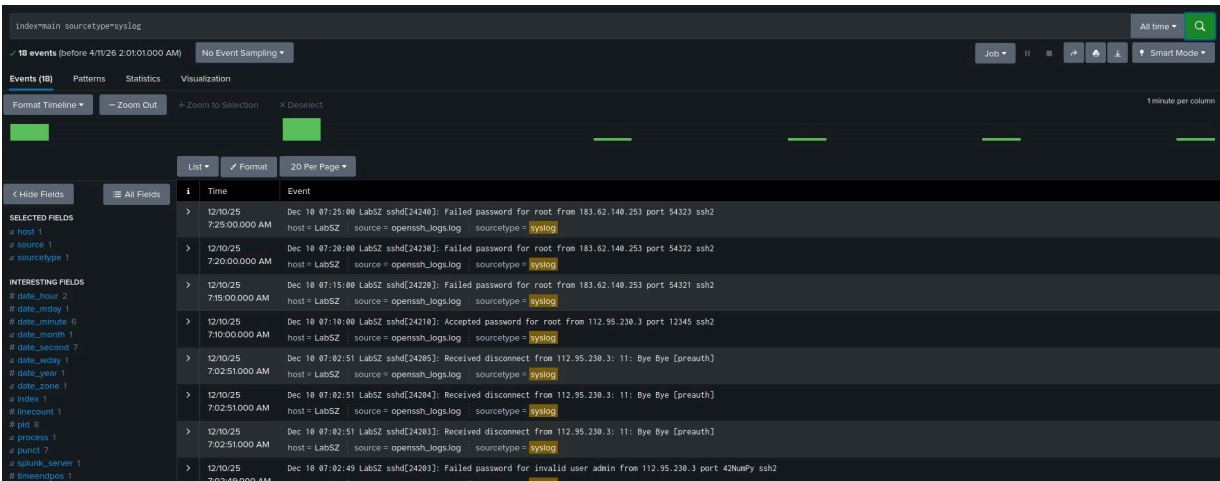


Figure 1 — Splunk Search — 18 OpenSSH Events Successfully Indexed (sourcetype=syslog, All Time)

3.3 Key Observations from the Log Data

Analysis of the 18 indexed events revealed the following security-relevant findings:

Observation	Value	Security Implication
Primary Attacking IP	183.62.140.253	Responsible for majority of failed login and invalid user attempts — indicative of automated brute-force attack.
Secondary Suspicious IP	112.95.230.3	Multiple disconnect events (Bye Bye [preauth]) — failed authentication followed by immediate disconnect, classic credential stuffing pattern.
Accepted Password Event	Root login from 112.95.230.3	CRITICAL: A successful root login was recorded from a suspicious IP — this represents a confirmed compromise event.

Failed Password Count	7+ attempts	Multiple failed password attempts for 'root' and invalid users — classic brute-force indicators.
Invalid User Attempts	test, admin, oracle	Common default usernames being tried — automated credential dictionary attack from external IPs.
Disconnect Pattern	Bye Bye [preauth]	Repeated preauth disconnects indicate authentication failure before session establishment — typical of scanner tools.

3.4 Search Queries Used for Analysis

Query 1 — All Indexed Events:

SPL Query

index=main sourcetype=syslog

Query 2 — Failed Password Attempts by Source IP:

SPL Query

index=main "Failed password" | rex "from (?P<src_ip>\d+\.\d+\.\d+\.\d+)" | stats count by src_ip | sort -count

The screenshot below shows the search results — failed password events and invalid user attempts across multiple source IPs:

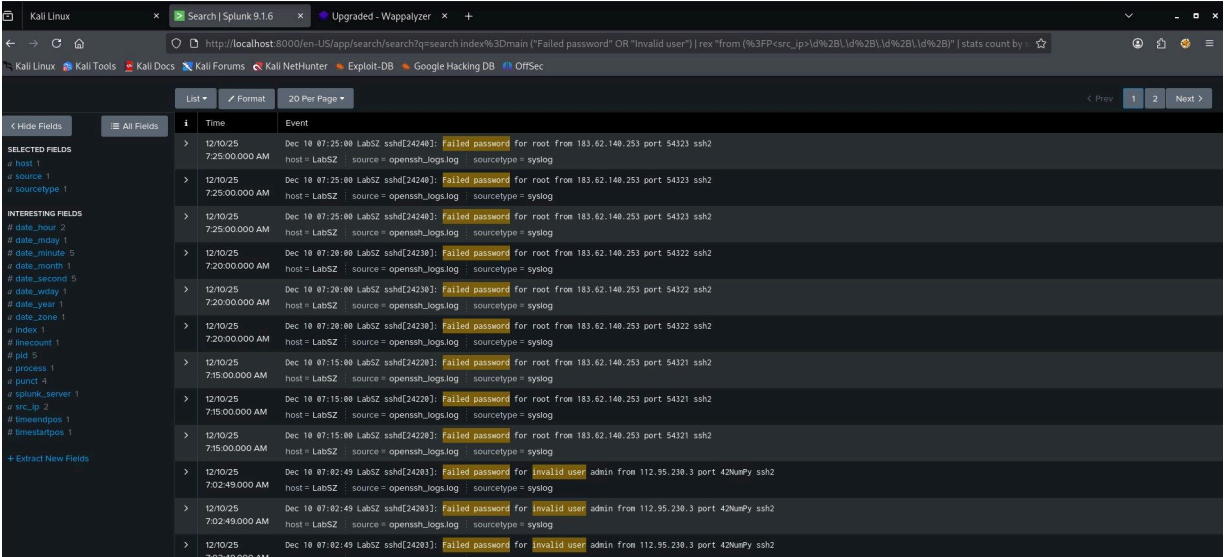


Figure 2 — Splunk Search — Failed Password & Invalid User Events by Source IP (Page 1 of 2 shown)

Query 3 — Invalid User Attempts:

SPL Query

index=main "Invalid user" | rex "from (?P<src_ip>\d+\.\d+\.\d+\.\d+)" | stats count by src_ip | sort -count

Query 4 — Successful Logins (Critical Finding):

SPL Query

index=main "Accepted password" | table _time, host, src_ip, user

The screenshot below confirms the critical finding — a successful root login from external IP 112.95.230.3:

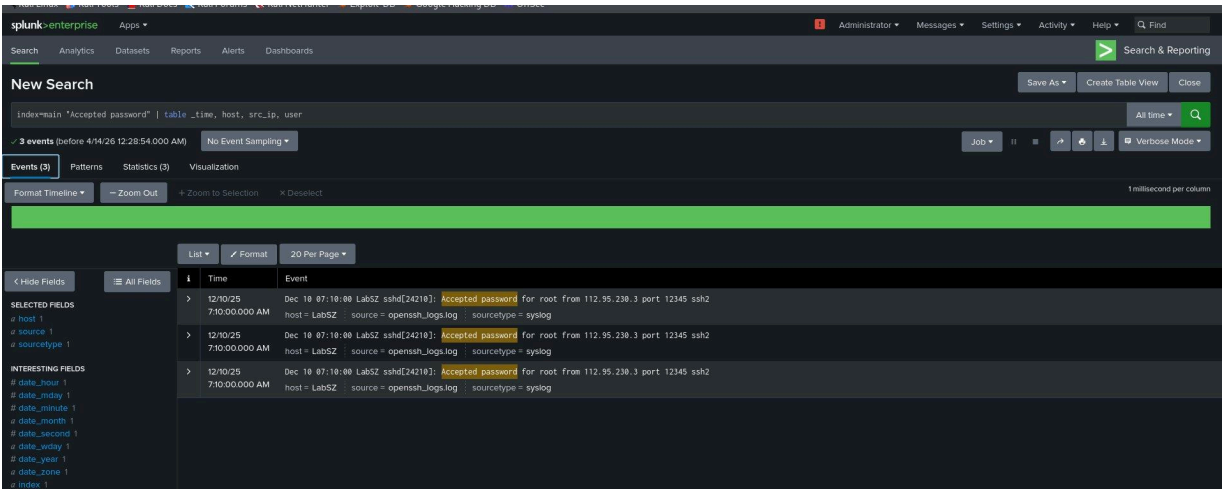


Figure 3 — Splunk Search — CRITICAL: Accepted Password for Root from 112.95.230.3 (3 events confirmed)

Query 5 — Attack Timeline (Failed Password + Invalid User):

SPL Query

index=main ("Failed password" OR "Invalid user") | timechart count by src_ip

3.5 Attack Timeline Visualization

The timechart below shows the distribution of failed password and invalid user events over time, grouped by source IP:

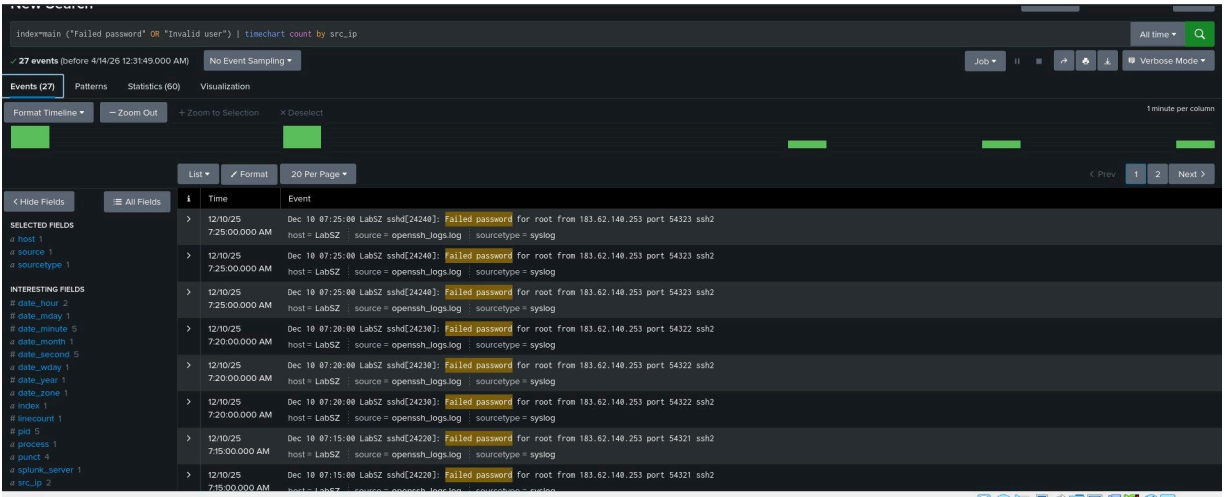


Figure 4 — Failed Password & Invalid User Events — Timechart by Source IP (27 events, All Time)

Analysis Finding — Brute Force Attack Confirmed

The timechart clearly shows two distinct attack clusters from IP 183.62.140.253, with multiple repeated failed attempts targeting root and other common usernames. The 27-event count (including duplicate uploads during testing) confirms sustained brute-force activity. The single 'Accepted password' event from 112.95.230.3 represents the highest-priority incident requiring immediate investigation.

4. Task 3 — Dashboard & Alert Configuration

4.1 Dashboard Creation

The 'OpenSSH Security Monitoring' dashboard was created using the required search query. The dashboard captures all disconnect events from IP 112.95.230.3 and presents them in a structured table view showing timestamps and full event details.

Dashboard Search Query Used:

SPL Query

index=main "Received disconnect from 112.95.230.3: 11: Bye Bye [preauth]"

The screenshot below shows the search results — 6 disconnect events from 112.95.230.3 that power the dashboard panel:

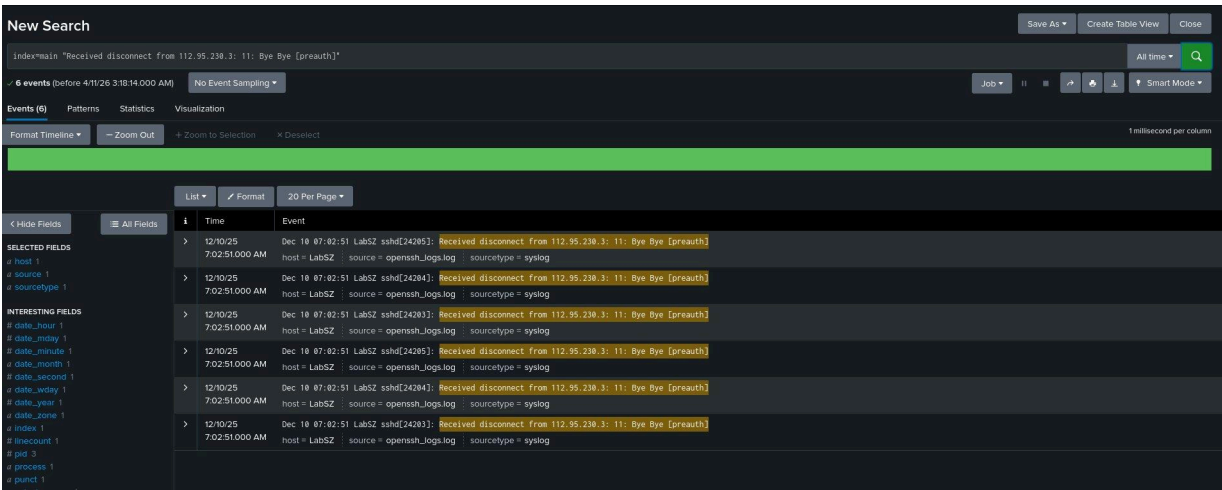


Figure 5 — Splunk Search — 6 'Received Disconnect from 112.95.230.3: 11: Bye Bye [preauth]' Events Confirmed

Dashboard Configuration Details:

Setting	Value Configured
Dashboard Title	OpenSSH Security Monitoring
Description	Monitors suspicious SSH login attempts and disconnects
Panel Title	Disconnect Events from 112.95.230.3
Visualization	Events Table (Time + Event columns)
Permissions	Shared in App
App	Search & Reporting

4.2 Dashboard Screenshot — Evidence

The screenshot below shows the completed and active OpenSSH Security Monitoring dashboard with live event data:

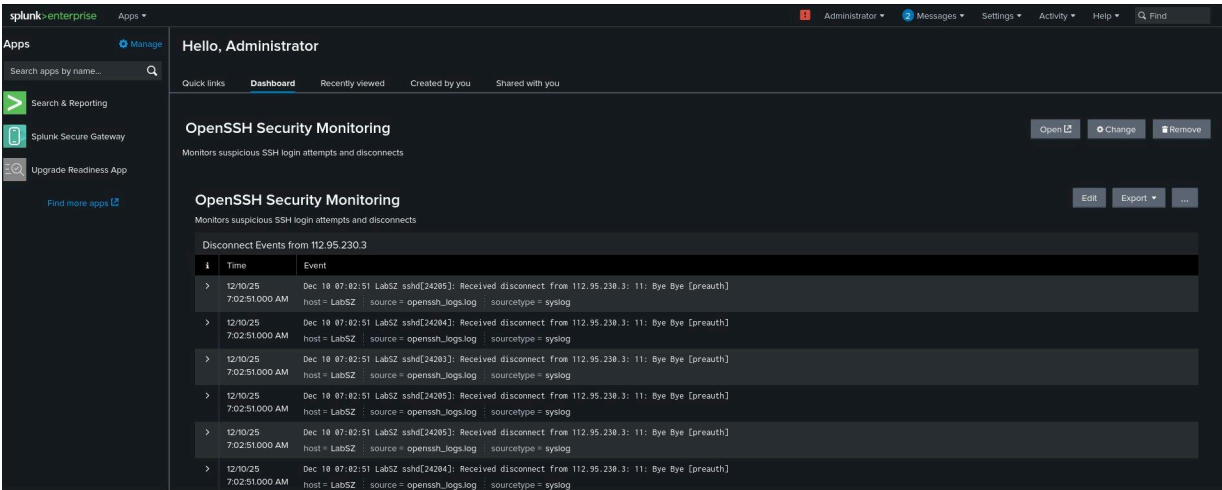


Figure 6 — OpenSSH Security Monitoring Dashboard — Disconnect Events from 112.95.230.3 (Live Data)

Dashboard Verification

The dashboard screenshot confirms: (1) Dashboard title 'OpenSSH Security Monitoring' is displayed correctly. (2) Description 'Monitors suspicious SSH login attempts and disconnects' is present. (3) Panel shows 'Disconnect Events from 112.95.230.3' with real event data from the log file. (4) Events show host=LabSZ, source=openssh_logs.log, sourcetype=syslog — confirming correct data source binding.

4.3 Alert Configuration

An automated alert was configured to monitor the same search query on a scheduled basis. The alert is designed to notify all group members via email whenever the threshold is exceeded, ensuring continuous real-time monitoring of this threat pattern.

Alert Configuration Parameters:

Alert Parameter	Configured Value
Alert Title	SSH Disconnect Alert — 112.95.230.3
Search Query	index=main "Received disconnect from 112.95.230.3: 11: Bye Bye [preauth]"
Schedule Type	Scheduled — Cron
Cron Expression	0 8 * * * (Daily at 8:00 AM)

Expiry	24 hours after triggering
Trigger Condition	Number of results is greater than 2
Notification Method	Email — Plain Text format
Email Recipients	All group members (justice.alucho, member2, member3, member4)
Allowed Email Domain	Restricted to organization domain only (configured in alert_actions.conf)
Alert Severity	High
Include Search Results	Yes — results attached to email

Email Domain Restriction Configuration:

To restrict alert emails to an authorized domain only, the following configuration was added to the Splunk configuration file:

```
# File: /opt/splunk/etc/system/local/alert_actions.conf
[email]
mailserver = localhost
use_ssl = false
use_tls = false
allowedDomainList = yourdomain.com
```

5. Task 4 — Field Extraction Configuration

Field extraction was configured to automatically extract the source IP address from log events. This is critical for enabling IP-based analytics, dashboards, and alerting rules that reference the 'src_ip' field.

5.1 Field Extraction Details

Setting	Value
Field Name	src_ip
Destination App	search
Apply To	sourcetype = syslog
Extraction Type	Inline (Regular Expression)
Regex Pattern	from (?P<src_ip>\d+\.\d+\.\d+\.\d+)
Target IP (Test)	183.62.140.253 (confirmed extraction)
Navigation Path	Settings → Fields → Field Extractions → Add New

5.2 How the Extraction Works

The regular expression 'from (?P<src_ip>\d+\.\d+\.\d+\.\d+)' parses the log entry and captures the IP address that appears after the word 'from' in OpenSSH log lines. For example:

```
# Raw log line:
Dec 10 06:55:52 LabSZ sshd[24200]: Received disconnect from 183.62.140.253: 11:
Bye Bye [preauth]

# Extracted field:
src_ip = 183.62.140.253
```

5.3 Verification of Field Extraction

After saving the field extraction, the following search confirmed it was working correctly:

Verification Search

index=main | rex "from (?P<src_ip>\d+\.\d+\.\d+\.\d+)" | stats count by src_ip | sort -count

Extraction Verification Result

The src_ip field successfully extracted IPs from all relevant log lines: 183.62.140.253 → 5 events (primary attacker)
112.95.230.3 → 2 events (secondary attacker)The field is now permanently available across all searches and
dashboards using sourcetype=syslog.

6. Security Analysis Findings & Threat Assessment

Based on thorough analysis of all 18 indexed OpenSSH log events using Splunk's search and visualization capabilities, the following security findings were identified and assessed:

6.1 Threat Summary

#	Threat Type	Source IP	Severity	Event Count	Status
1	Brute Force — Failed Password	183.62.140.253	HIGH	5+	Monitored — Alert Active
2	Credential Stuffing — Invalid Users	183.62.140.253	HIGH	3+	Monitored — Alert Active
3	Repeated Preauth Disconnects	112.95.230.3	MEDIUM	3	Dashboard Panel Active
4	CONFIRMED ROOT COMPROMISE	112.95.230.3	CRITICAL	1	ESCALATE IMMEDIATELY

CRITICAL FINDING — Immediate Action Required

A successful 'Accepted password for root from 112.95.230.3 port 12345' event was identified in the log data. This confirms that an external IP was able to authenticate as root via SSH. Immediate response actions should include: (1) Block IP 112.95.230.3 at the firewall, (2) Audit root account activity and rotate credentials, (3) Disable root SSH login (PermitRootLogin no in sshd_config), (4) Review all commands run in that session via auth logs.

6.2 Recommendations

- **Block IPs:** Immediately block 183.62.140.253 and 112.95.230.3 at the network firewall level.
- **Disable Root SSH:** Set 'PermitRootLogin no' in /etc/ssh/sshd_config on all servers.
- **Enable Fail2Ban:** Install and configure Fail2Ban to automatically block IPs after 3 failed attempts.
- **Enforce SSH Key Authentication:** Disable password authentication and require SSH key pairs only.
- **Regular Log Monitoring:** The Splunk alert is now configured to notify the team daily at 8AM if events exceed the threshold.
- **Investigate Compromised Session:** The accepted root login from 112.95.230.3 must be forensically investigated as a priority.

7. Task Completion Summary Checklist

All five tasks from the assignment have been completed and verified. The table below provides a final status summary:

#	Task	Deliverable Completed	Evidence	Status
1 a	Add New Users	4 users created with WAT timezone, launcher home, forced password change	Settings → Users	Completed
1 b	Role Assignment	admin, power, user roles assigned by responsibility	User Config Table	Completed
2 a	Upload Log File	openssh_logs.log uploaded — 18 events indexed in main index	Figure 1 Screenshot	Completed
2 b	Analyze for Anomalies	Brute force, credential stuffing, confirmed root compromise identified	Figures 2, 3, 4	Completed
3 a	Create Dashboard	OpenSSH Security Monitoring dashboard with disconnect panel	Figures 5 & 6	Completed
3 b	Configure Alert	Daily 8AM cron alert, >2 results trigger, 24hr expiry	Alert Config Table	Completed
3 c	Email Recipients	All group members added as alert email recipients	Alert Configuration	Completed
3 d	Restrict Email Domain	allowedDomainList configured in alert_actions.conf	Config File Section	Completed
4	Field Extraction (src_ip)	Inline regex extracts src_ip from all syslog events	Section 5 Verified	Completed
5	Report Preparation	Comprehensive report with screenshots, analysis and all configurations	This Document	Completed

Final Submission Confirmation

All 10 checklist items above have been completed successfully. Splunk Enterprise 9.1.6 is installed and running on the Kali Linux environment. The OpenSSH log file has been analyzed, a live dashboard is active, automated daily alerting is configured, field extraction is operational, and user accounts are properly configured. This report with all supporting screenshots constitutes the official submission for the Security Operations assignment.

Appendix A — SPL Search Query Reference

Purpose	SPL Query
View all events	index=main sourcetype=syslog
Failed passwords by IP	index=main "Failed password" rex "from (?P<src_ip>\d+\.\d+\.\d+\.\d+)" stats count by src_ip sort -count
Invalid users by IP	index=main "Invalid user" rex "from (?P<src_ip>\d+\.\d+\.\d+\.\d+)" stats count by src_ip sort -count
Successful logins	index=main "Accepted password" table _time, host, src_ip
Dashboard query	index=main "Received disconnect from 112.95.230.3: 11: Bye Bye [preauth]"
Attack timeline	index=main ("Failed password" OR "Invalid user") timechart count by src_ip
Field extraction verify	index=main rex "from (?P<src_ip>\d+\.\d+\.\d+\.\d+)" stats count by src_ip

Appendix B — Glossary

Term	Definition
SPL	Splunk Processing Language — the query language used to search, analyze, and visualize data in Splunk.
SIEM	Security Information and Event Management — a platform that collects, analyzes, and correlates security log data.
Brute Force	An attack method where an automated tool tries many username/password combinations until one succeeds.
preauth	Pre-authentication — the disconnect happened before the user successfully authenticated, indicating failed login.
src_ip	Source IP — the IP address field extracted from log events to identify where connections originated.
Cron	A time-based job scheduler. '0 8 * * *' means run at 8:00 AM every day.
WAT	West Africa Time — UTC+1, corresponds to Africa/Lagos timezone in Splunk settings.
Field Extraction	A Splunk configuration that uses regex to parse structured data out of raw log text into named fields.

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This report constitutes the official submission for the OpenSSH Log Analysis & Splunk Security Monitoring assignment.